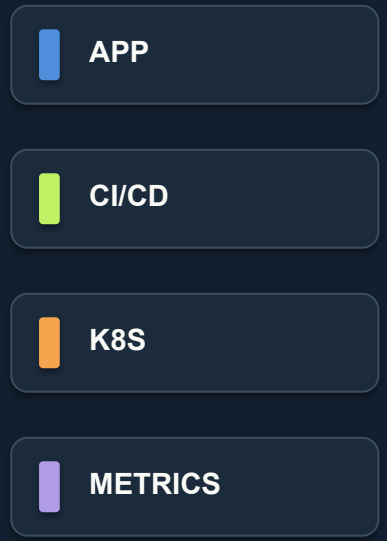
DEVOPS CAPSTONE PROJECT

CampusTask

Student Assignment & Project Tracker

From application code to automated delivery, Kubernetes, infrastructure as code, and observability.



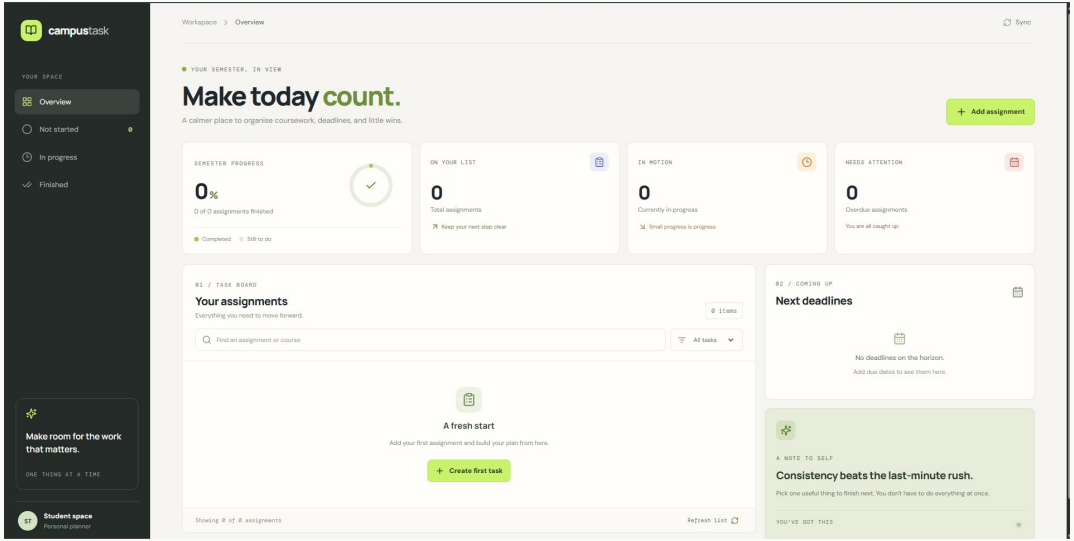
React • FastAPI • PostgreSQL • Docker • GitHub Actions • Kubernetes • Terraform • Prometheus • Grafana

GitHub: github.com/Nirbhay-kr24/CampusTask

A calmer place to manage coursework

CampusTask is a student planner for organizing assignments, tracking progress, and keeping deadlines visible.

- Create, list, update, search, and delete tasks
- Track progress and view task statistics
- Expose health, readiness, and metrics endpoints
- Demonstrate a DevOps delivery workflow



A practical, modular stack

Frontend

React + Vite

Student-facing dashboard

API

Python + FastAPI

REST endpoints and Swagger docs

Data

PostgreSQL

SQLAlchemy + Alembic

Containers

Docker + Compose

Repeatable local environment

Delivery

GitHub Actions

Tests • CodeQL • Trivy • image publishing

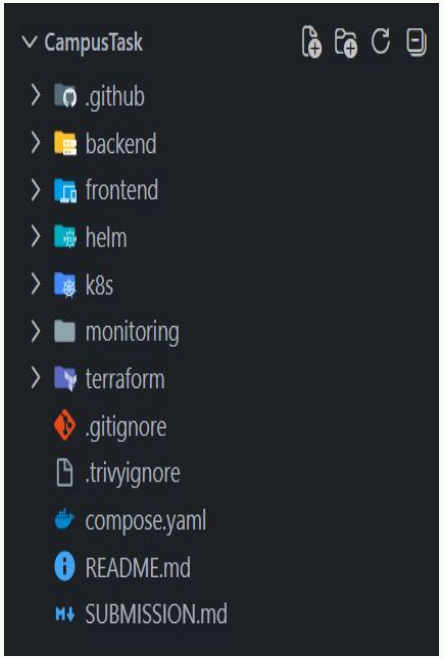
Operations

Kubernetes + Helm

Terraform • Prometheus • Grafana

A modular design keeps UI, API, persistence, and operations independently manageable.

Application architecture

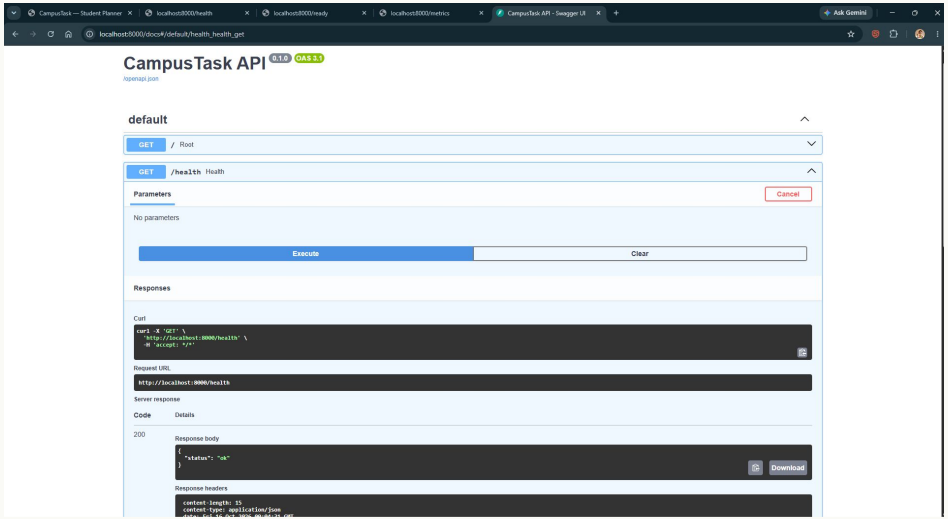


Student
Browser / UI

React + Vite
Frontend

FastAPI
REST API

PostgreSQL
Persistent data



/health • /ready
Availability checks

/metrics
Prometheus scrape

Prometheus → Grafana
Collection + dashboards

Automated delivery with security gates

01 Push — GitHub commit

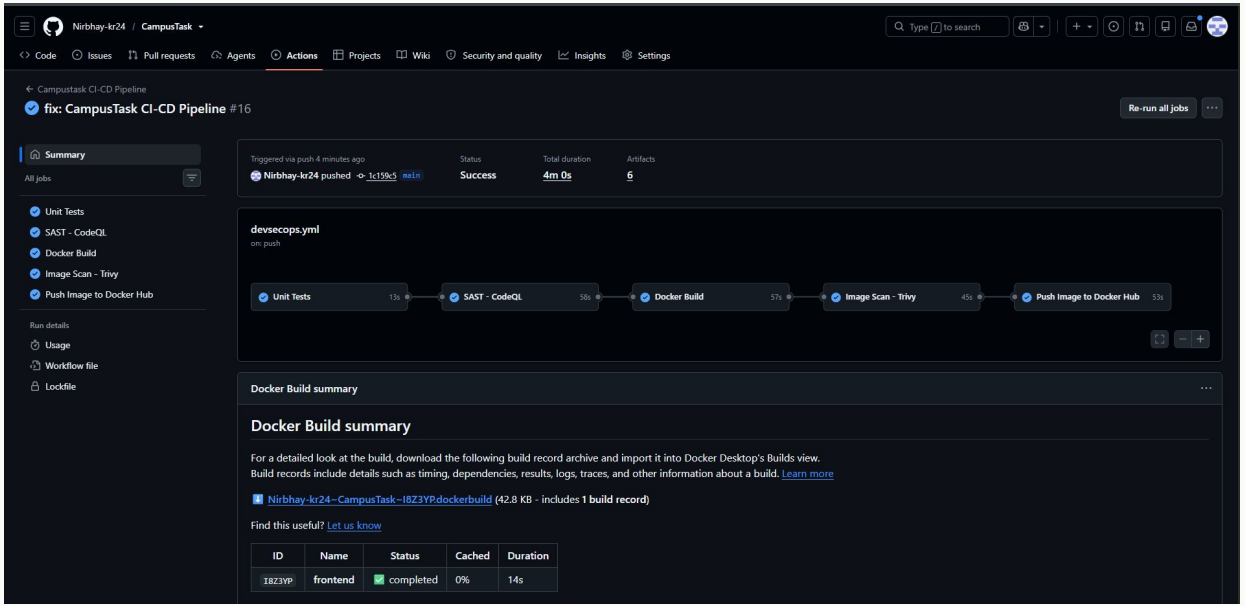
02 Test — Pytest

03 Analyze — CodeQL

04 Build — Docker images

05 Scan — Trivy

06 Publish — Docker Hub



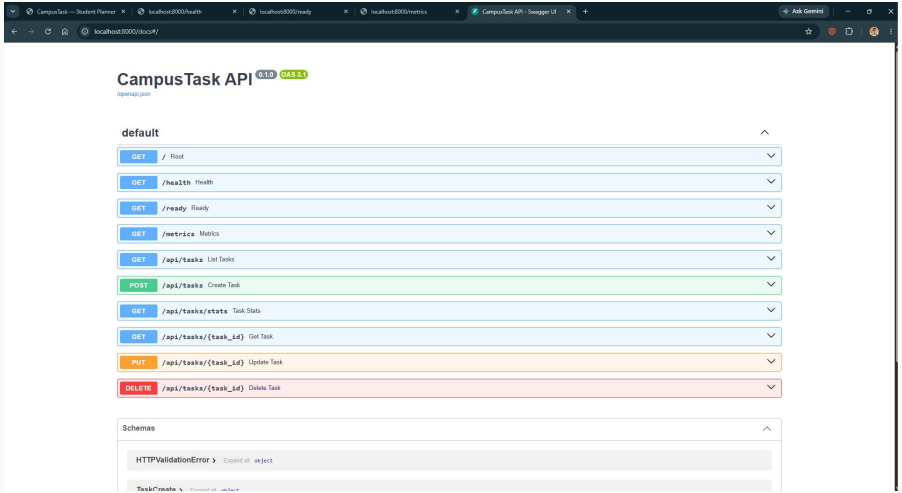
Evidence: successful workflow

Screenshot shows tests, CodeQL, Docker build, Trivy scan, and image push jobs succeeding.

Why it matters

Repeatable checks catch issues earlier and reduce manual release steps.

Run, inspect, and test the app

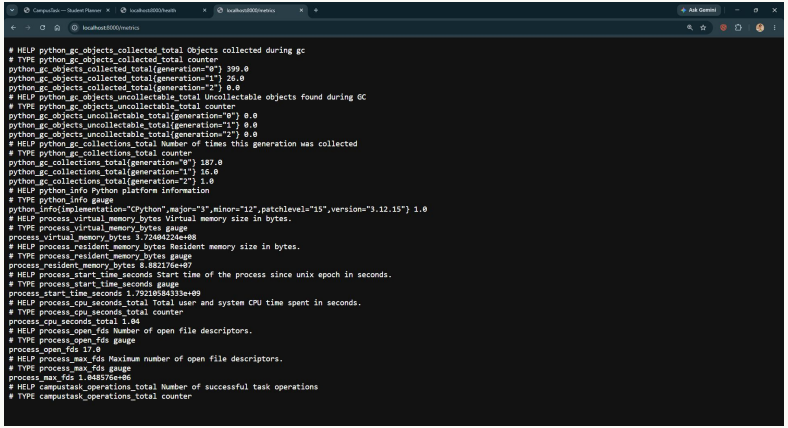



```
nirbhay@NirbhayOMEN:/mnt/c/Users/nirbh/Downloads/CampusTask$ docker run --rm campustask-backend:ci pytest
-v -p no:cacheprovider
============================= test session starts ==============================
platform linux -- Python 3.12.15, pytest-8.3.5, pluggy-1.7.0 -- /usr/local/bin/python3.12
rootdir: /app
plugins: anyio-4.15.1
collecting ... collected 9 items

tests/test_tasks.py::test_health PASSED                                  [ 11%]
tests/test_tasks.py::test_create PASSED                                  [ 22%]
tests/test_tasks.py::test_list PASSED                                    [ 33%]
tests/test_tasks.py::test_missing_task PASSED                            [ 44%]
tests/test_tasks.py::test_update PASSED                                  [ 55%]
tests/test_tasks.py::test_delete PASSED                                  [ 66%]
tests/test_tasks.py::test_stats PASSED                                   [ 77%]
tests/test_tasks.py::test_search PASSED                                  [ 88%]
tests/test_tasks.py::test_empty_title_rejected PASSED                    [100%]

============================== warnings summary ================================
../usr/local/lib/python3.12/site-packages/starlette/testclient.py:37
  /usr/local/lib/python3.12/site-packages/starlette/testclient.py:37: DeprecationWarning: The anyio.abc.Bl
ockingPortal alias is deprecated, use anyio.from_thread.BlockingPortal instead.
    _PortalFactoryType = typing.Callable[[], typing.ContextManager[anyio.abc.BlockingPortal]]

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
======================== 9 passed, 1 warning in 0.14s ==========================
```

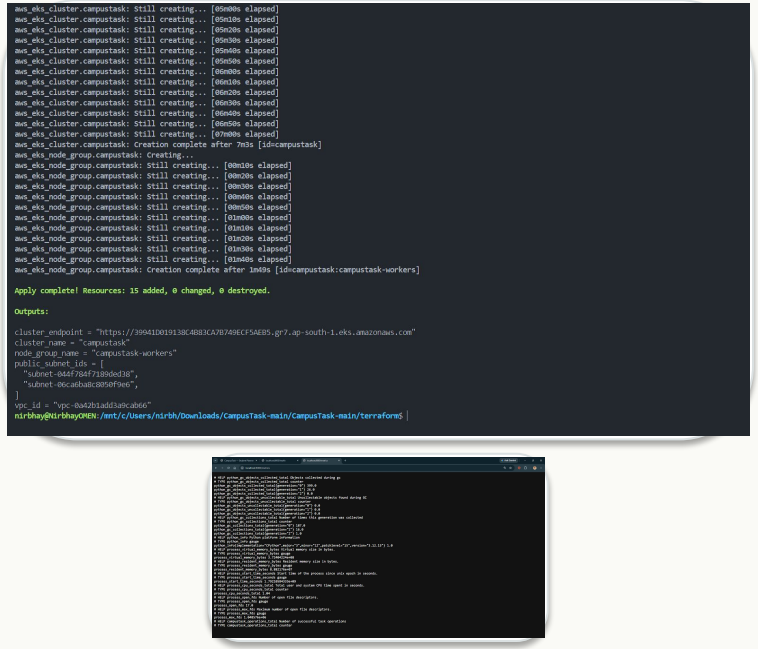


```
# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 399.0
python_gc_objects_collected_total{generation="1"} 26.0
python_gc_objects_collected_total{generation="2"} 0.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 187.0
python_gc_collections_total{generation="1"} 16.0
python_gc_collections_total{generation="2"} 1.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="12",patchlevel="15",version="3.12.15"} 1.0
# HELP process_virtual_memory_bytes Virtual memory size in bytes.
# TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 3.72404224e+08
# HELP process_resident_memory_bytes Resident memory size in bytes.
# TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 8.882176e+07
# HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
# TYPE process_start_time_seconds gauge
process_start_time_seconds 1.792108543138e+09
# HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
# TYPE process_cpu_seconds_total counter
process_cpu_seconds_total 1.04
# HELP process_open_fds Number of open file descriptors.
# TYPE process_open_fds gauge
process_open_fds 17.0
# HELP process_max_fds Maximum number of open file descriptors.
# TYPE process_max_fds gauge
process_max_fds 1.048576e+06
# HELP campustask_operations_total Number of successful task operations
# TYPE campustask_operations_total counter
```

Swagger documents the REST API. Tests recorded 9 passed; /metrics returned Prometheus-format data.

Kubernetes deployment and routing

- Deployments manage frontend and backend replicas
- PostgreSQL runs as a dedicated workload with storage configuration
- Services provide stable networking between components
- Helm chart groups reusable manifests and values
- Ingress routes HTTP traffic into the application

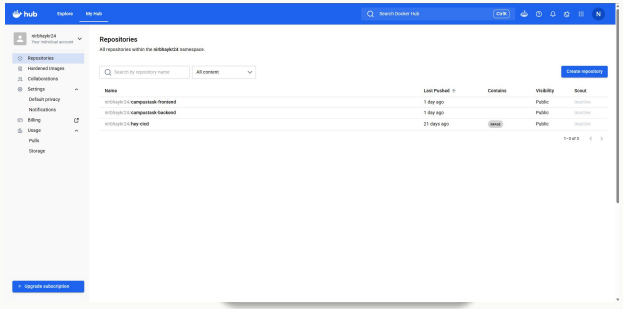


Evidence: running workloads and configured ingress resource

Infrastructure as code with Terraform

Terraform made the infrastructure lifecycle repeatable and reviewable.

- Initialize providers and validate configuration
- Review the plan before applying changes
- Provisioned an AWS EKS cluster and worker node group
- Destroyed resources after testing to control ongoing cost




```
nirbhay@NirbhayOMEN:/mnt/c/Users/nirbh/Downloads/CampusTask-main/CampusTask-main$ ls
README.md  backend  compose.yaml  frontend  helm  k8s  monitoring  terraform
nirbhay@NirbhayOMEN:/mnt/c/Users/nirbh/Downloads/CampusTask-main/CampusTask-main$ cd terraform
nirbhay@NirbhayOMEN:/mnt/c/Users/nirbh/Downloads/CampusTask-main/CampusTask-main/terraform$ terraform init
Initializing the backend...

Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/helm from the dependency lock file
- Reusing previous version of hashicorp/kubernetes from the dependency lock file
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
- Installing hashicorp/helm v3.3.0...
- Installed hashicorp/helm v3.3.0 (signed by HashiCorp)
- Installing hashicorp/kubernetes v2.38.0...
- Installed hashicorp/kubernetes v2.38.0 (signed by HashiCorp)

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
nirbhay@NirbhayOMEN:/mnt/c/Users/nirbh/Downloads/CampusTask-main/CampusTask-main/terraform$ terraform validate
Success! The configuration is valid.

nirbhay@NirbhayOMEN:/mnt/c/Users/nirbh/Downloads/CampusTask-main/CampusTask-main/terraform$ terraform plan

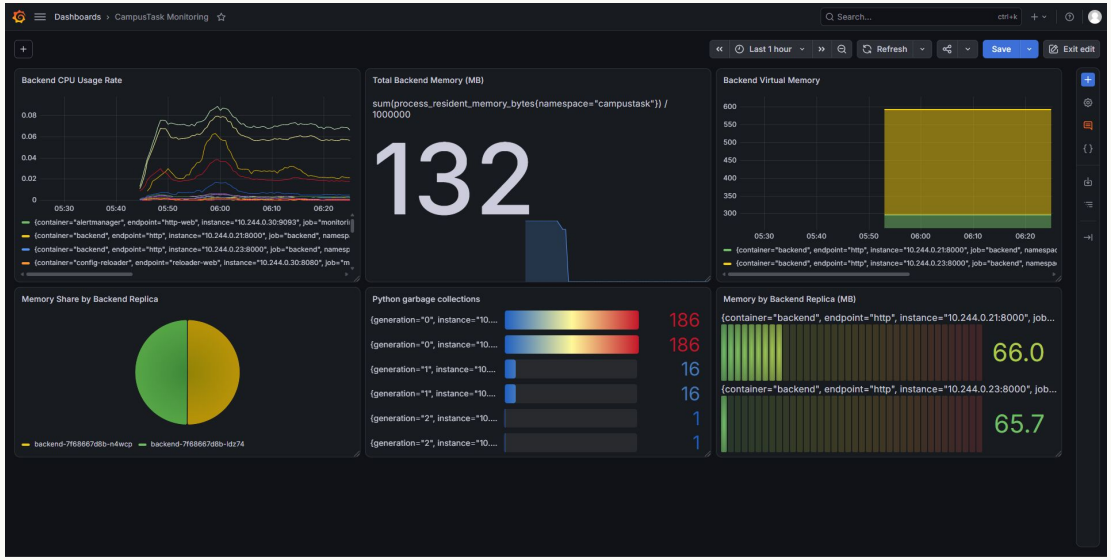
Terraform used the selected providers to generate the following execution plan. Resource actions are
indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_eks_cluster.campustask will be created
  + resource "aws_eks_cluster" "campustask" {
      + arn                           = (known after apply)
```

Lifecycle evidence: plan → apply (15 resources) → destroy (15 resources). Verify AWS cleanup with valid credentials.

Observe the system, not just the UI



Prometheus

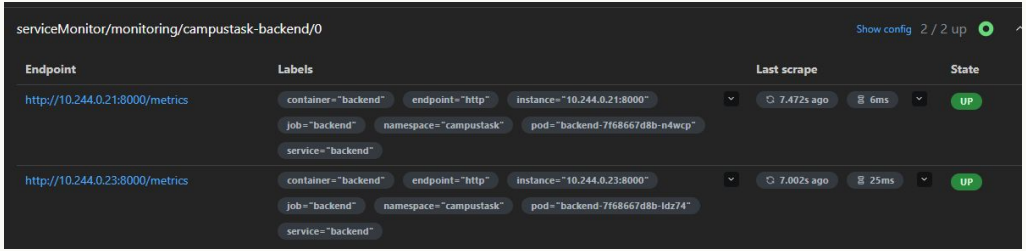
Scrapes application and platform metrics from configured targets.

Grafana

Dashboards visualize CPU, memory, and process metrics.

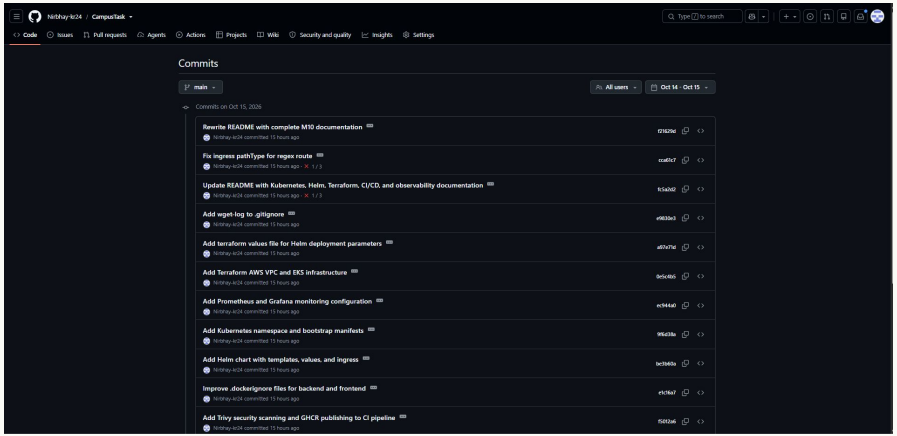
Health endpoints

/health and /ready support health and dependency checks.



Images, versions, and project history

```
nirbhay@NirbhayOMEN:/mnt/c/Users/nirbh/Downloads/CampusTask-main/CampusTask-main/terraform$ kubectl get pods -A
NAMESPACE       NAME                                                READY   STATUS      RESTARTS      AGE
campustask      backend-7f68667d8b-ldz74                            1/1     Running     0             14h
campustask      backend-7f68667d8b-n4wcp                            1/1     Running     0             14h
campustask      frontend-8d944d85-9hskk                             1/1     Running     0             15h
campustask      frontend-8d944d85-w8dl2                             1/1     Running     0             14h
campustask      postgres-56c674cc47-2j4fm                           1/1     Running     0             15h
ingress-nginx   ingress-nginx-admission-create-w7gdb                0/1     Completed   0             38h
ingress-nginx   ingress-nginx-admission-patch-49t9k                 0/1     Completed   0             38h
ingress-nginx   ingress-nginx-controller-d7cd8c989-8pdn4            1/1     Running     0             38h
kube-system     coredns-559f6c778d-97kkn                            1/1     Running     1 (18h ago)   38h
kube-system     coredns-559f6c778d-lbk42                            1/1     Running     1 (18h ago)   38h
kube-system     etcd-minikube                                       1/1     Running     1 (18h ago)   38h
kube-system     kindnet-wp57j                                       1/1     Running     1 (18h ago)   38h
kube-system     kube-apiserver-minikube                             1/1     Running     1 (18h ago)   38h
kube-system     kube-controller-manager-minikube                    1/1     Running     1 (18h ago)   38h
kube-system     kube-proxy-4pdgj                                    1/1     Running     1 (18h ago)   38h
kube-system     kube-scheduler-minikube                             1/1     Running     1 (18h ago)   38h
kube-system     storage-provisioner                                 1/1     Running     2 (18h ago)   38h
monitoring      alertmanager-monitoring-kube-prometheus-alertmanager-0   2/2     Running     0             7h7m
monitoring      monitoring-grafana-5c798557cf-q8x5b                 3/3     Running     0             7h5m
monitoring      monitoring-kube-prometheus-operator-b7bf5f7d8-q6j8h   1/1     Running     0             7h8m
monitoring      monitoring-kube-state-metrics-78fd56fc4b-9qxfg      1/1     Running     0             7h8m
monitoring      monitoring-prometheus-node-exporter-sm4jr           1/1     Running     0             7h8m
monitoring      prometheus-monitoring-kube-prometheus-prometheus-0  2/2     Running     0             7h7m
```



Container registry

Frontend and backend image repositories are visible in Docker Hub.

Versioned work

Commit history captures CI/CD, Helm, Terraform, monitoring, and documentation changes.

What this project taught me

Automate	Repeat tests, builds, and security checks in CI.
Containerize	Package services consistently with Docker.
Orchestrate	Use Kubernetes, Helm, Services, and Ingress.
Observe	Use health endpoints, Prometheus, and Grafana.
Manage safely	Use Terraform and verify infrastructure teardown.

Next: deployment approvals, actionable alerts, and documented recovery procedures.